

DATA-260 Homework 3, Poushali Purkayastha

Configuration

Value	Result
SID4	0571
PORT_BASE	$8000 + (571 \bmod 900) = 8571$
PREFIX	s0571
SEED	571
VERIFY_SEED	260571
DOMAIN_ID	$571 \bmod 8 = 3$, Grocery supply and recall notices
Hardware	Intel Core i5-1135G7 (4 cores, 8 threads), 15.8 GB RAM, Intel Iris Xe Graphics, Windows 11 Home, CPU only
Local model	sentence-transformers/all-MiniLM-L6-v2 (384-dimensional sentence embeddings, pulled from Hugging Face, run on CPU). This homework is retrieval only, so no LLM is called.
Software	Python 3.12.10, fastapi 0.141.1, starlette 1.0.0, uvicorn 0.46.0, jinja2 3.1.6, itsdangerous 2.2.0, llama-index 0.14.24, llama-index-embeddings-huggingface 0.8.0, sentence-transformers 6.0.1, torch 2.14.0, numpy 2.4.4, pandas 3.0.2, faiss-cpu 1.15.0
Repository	https://github.com/Poushali0202/data260-0571 (collaborators Sbnikitha and supriyaselvangesan)
Tagged commit	hw3 = 86892f0b7f5ee62e12b05dd15fa613f632148b48

The HW1 and HW2 code is extended in place. `app.py` keeps the HW2 notice API and now adds Starlette's `SessionMiddleware` and the new router `routers/auth.py` with the Jinja templates in `templates/`; the HW2 notice page moved from `/` to `/notices` so that `/` can be the home page asked for here. New root-level files: `fetch_corpus.py`, `corpus/` (21 domain documents), `chunking_compare.py`, `summarize_retrieval.py`, `verify_hw03.py`, and the `verify-hw03` target in the Makefile. Everything for this homework is in `reports/hw03/`: `RUN_LOG.txt`, `raw/`, `METRICS.md`, `AI_USE.md`, `README.md` (run instructions), `verification.json`, `SOURCES.md`, `CORPUS_MANIFEST.json`, `questions.yaml`, and `screenshots/`.

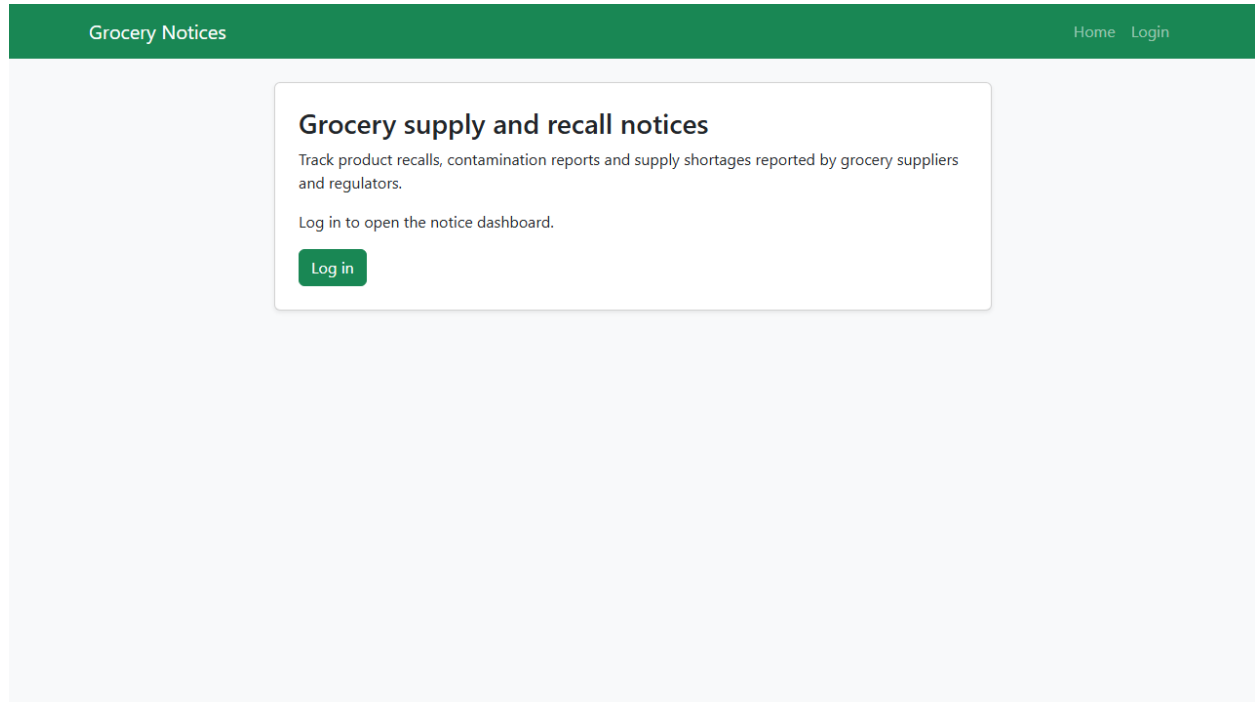
Part 1: FastAPI login system

Routes

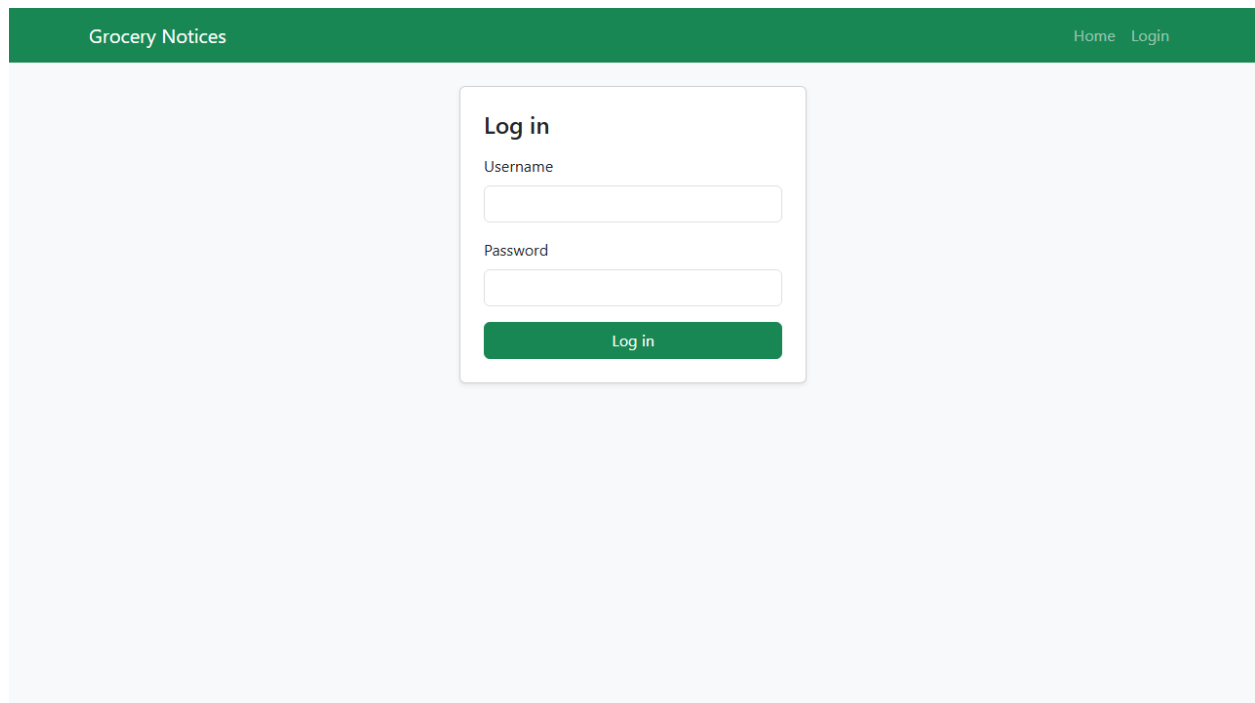
All four routes live in `routers/auth.py` and are registered with `app.include_router(auth_router)`. The home page shows a welcome message for the grocery notice application with a login link, or the dashboard and logout links once a user is logged in. The login page validates the form against a small user table, redirects to `/dashboard` on success, and re-renders the form with a Bootstrap `alert-danger` (status 401) when the credentials are wrong. The dashboard greets the user by name, links to the notice list and to logout, and redirects anonymous or

expired sessions to `/login`. Logout removes the session id from the server-side set, clears the session, and redirects to `/`.

Home page (`/`) while logged out:



Login page (`/login`):



Login page after a wrong password (Bootstrap alert):

Grocery Notices

Home Login

Log in

Invalid username or password.

Username

Password

Log in

Dashboard (/dashboard) after logging in as inspector:

Grocery Notices

Home Dashboard Logout

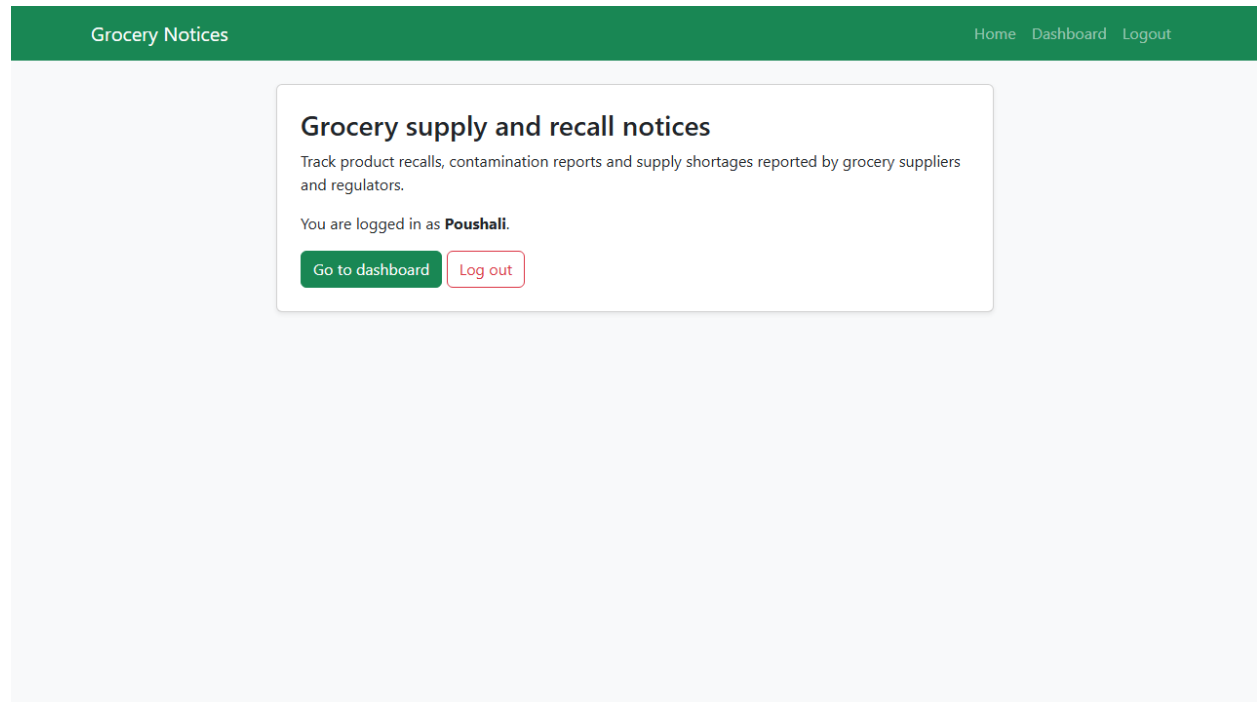
Welcome, Poushali

You are signed in as **inspector**. The notice list from the earlier homeworks is one click away.

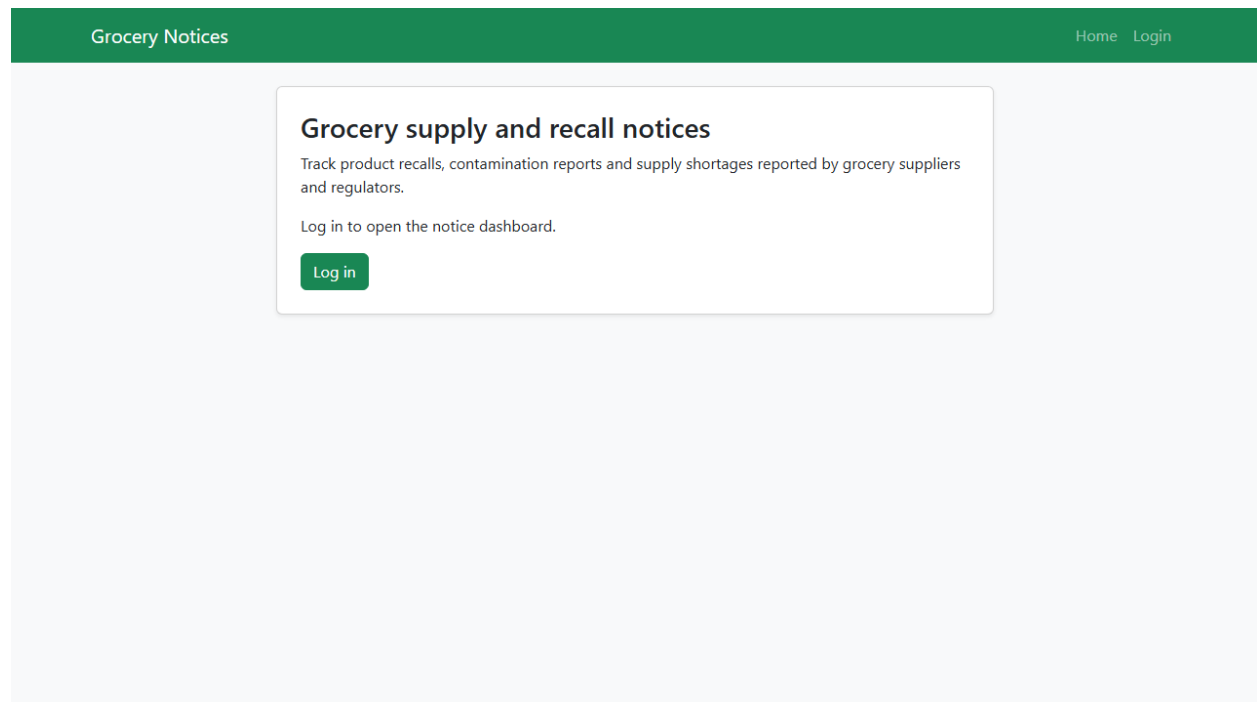
Open the notice list

Log out

Home page while logged in (dashboard and logout links):



/logout destroys the session and lands on the home page again:



Opening /dashboard after logout redirects to /login:

Log in

Username

Password

Log in

Session management

`app.py` installs the middleware with a secure cookie and a Max-Age equal to the idle timeout:

```
app.add_middleware(
    SessionMiddleware,
    secret_key=os.getenv("SECRET_KEY", "replace-me-s0571"),
    max_age=IDLE_TIMEOUT,
    same_site="lax",
    https_only=True,
)
app.include_router(auth_router)
```

Login stores the user (username and display name), a random session id and a `last_seen` timestamp in the session. The Set-Cookie header returned by a successful login carries all three attributes, `httponly`, `samesite=lax` and `secure` (captured with curl, full output in `RUN_LOG.txt`):

```
$ curl -i -s -X POST -d "username=inspector&password=recall2026" http://127.0.0.1:8571/login
HTTP/1.1 302 Found
location: /dashboard
vary: Cookie
set-cookie:
session=eyJpZCI6ICZ0Tk0NjYzNWFlZmFlNTVjZDRhOTFjNjlmMjQwZjQ4NyIsICJ1c2VyIjogeyJ1c2VybmFtZSI6ICJpbnNw
ZWNOb3IiLCBibmFtZSI6ICJQb3VzaGFsaSJ9LCAibGFzZDF9ZWVuIjogMTc4OTk3NTMxNC4wMzE4MDc3fQ==.arDbEg.dwcM-
22byFEEfxg5ChwM0WBfS0Q; path=/; Max-Age=20; httponly; samesite=lax; secure
```

Idle timeout: `current_user()` compares the stored `last_seen` with the current time and clears the session when the gap is longer than `IDLE_TIMEOUT` (600 seconds by default, `IDLE_TIMEOUT_SECONDS` in the environment). Every request by a logged-in user refreshes `last_seen`, which also re-signs the cookie, so the Max-Age of the cookie slides along with the activity. The server also keeps a set of active session ids: logout removes the id, so a copy of a logged-out cookie is refused even though its signature is still valid. The proof below was recorded against a server started with `IDLE_TIMEOUT_SECONDS=20`. The same cookie is sent to `/dashboard`

after logging out, and again after 22 seconds of inactivity; both attempts are redirected to /login:

```
$ curl -i -s -b "$COOKIE" http://127.0.0.1:8571/logout
HTTP/1.1 302 Found
location: /
set-cookie: session=null; path=/; expires=Thu, 01 Jan 1970 00:00:00 GMT; httponly; samesite=lax; secure

$ curl -i -s -b "$COOKIE" http://127.0.0.1:8571/dashboard (same cookie replayed after logout)
HTTP/1.1 302 Found
location: /login

$ date; curl -i -s -b "$COOKIE" http://127.0.0.1:8571/dashboard | head -1 (fresh login)
2026-09-21T00:21:54-07:00
HTTP/1.1 200 OK
$ sleep 22
$ date; curl -i -s -b "$COOKIE" http://127.0.0.1:8571/dashboard (same cookie after 22 s idle)
2026-09-21T00:22:16-07:00
HTTP/1.1 302 Found
location: /login
```

```
TERMINAL powershell
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> curl.exe -i -s -X POST -d "username=inspector&password=wrong" http://127.0.0.1:8571/login | Select-Object -First 1
HTTP/1.1 401 Unauthorized
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> $resp = curl.exe -i -s -X POST -d "username=inspector&password=recall2026" http://127.0.0.1:8571/login; $resp
HTTP/1.1 302 Found
date: Mon, 21 Sep 2026 07:54:16 GMT
server: uvicorn
content-length: 0
location: /dashboard
vary: Cookie
set-cookie: session=eyJpZCI6ICJlMMZmNjAxZWUzN2VlZDlmOTgxZjU3OGVmZDhmN2U3YiIsICJlc2VyIjoeyJ1c2VybmFtZSI6ICJpbmNwZWNoY3IiLCJlbmFtZSI6ICJQb3VzaGFsaS39LCAibGFzZD9zZWUiJmFtZC40Tk3NzI1Ni4xMjk0OTczfQ==.arDqA.V8Y7QNXEvM7BP4xh77dPBDSqpsE; path=/; Max-Age=20; httponly; samesite=lax; secure
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> $cookie = ($resp | Select-String "^set-cookie: (session=[^;]+)").Matches[0].Groups[1].Value
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> curl.exe -i -s -b $cookie http://127.0.0.1:8571/dashboard | Select-Object -First 1
HTTP/1.1 200 OK
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> curl.exe -i -s -b $cookie http://127.0.0.1:8571/logout | Select-Object -First 1
HTTP/1.1 302 Found
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> curl.exe -i -s -b $cookie http://127.0.0.1:8571/dashboard | Select-Object -First 1
HTTP/1.1 302 Found
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> $resp = curl.exe -i -s -X POST -d "username=inspector&password=recall2026" http://127.0.0.1:8571/login; $cookie = ($resp | Select-String "^set-cookie: (session=[^;]+)").Matches[0].Groups[1].Value
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> Get-Date -Format s; curl.exe -i -s -b $cookie http://127.0.0.1:8571/dashboard | Select-Object -First 1; Start-Sleep 22
2026-09-21T00:54:16
HTTP/1.1 200 OK
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> Get-Date -Format s; curl.exe -i -s -b $cookie http://127.0.0.1:8571/dashboard | Select-Object -First 5
2026-09-21T00:54:38
HTTP/1.1 302 Found
date: Mon, 21 Sep 2026 07:54:37 GMT
server: uvicorn
content-length: 0
location: /login
```

Styling with Bootstrap

Every page extends `templates/base.html`, which loads Bootstrap 5.3 from the CDN and renders a navbar with a navbar - toggler that collapses below the md breakpoint. Content sits in card components inside a centred row/col grid, actions are `btn btn-success` and `btn-outline-danger` buttons, and the login error is an `alert alert-danger`. At 375px the navbar collapses into the toggler menu and the card fills the width:

Grocery Notices

Home

Login

Log in

Username

Password

Log in

routers/auth.py

```
import os
import secrets
import time
from pathlib import Path

from fastapi import APIRouter, Form, Request
from fastapi.responses import RedirectResponse
from fastapi.templating import Jinja2Templates
router = APIRouter()
```

```

templates = Jinja2Templates(directory=Path(__file__).resolve().parent.parent / "templates")

USERS = {
    "inspector": {"password": "recall2026", "name": "Poushali"},
    "manager": {"password": "shelf2026", "name": "Store Manager"},
}
IDLE_TIMEOUT = int(os.getenv("IDLE_TIMEOUT_SECONDS", "600"))
ACTIVE_SESSIONS = set()

def current_user(request: Request):
    user = request.session.get("user")
    session_id = request.session.get("id")
    idle = time.time() - request.session.get("last_seen", 0)
    if not user or session_id not in ACTIVE_SESSIONS or idle > IDLE_TIMEOUT:
        ACTIVE_SESSIONS.discard(session_id)
        request.session.clear()
        return None
    request.session["last_seen"] = time.time()
    return user

@router.get("/")
def home(request: Request):
    return templates.TemplateResponse(request, "index.html", {"user": current_user(request)})

@router.get("/login")
def login_page(request: Request):
    return templates.TemplateResponse(request, "login.html", {"user": current_user(request),
"error": None})

@router.post("/login")
def login(request: Request, username: str = Form(), password: str = Form()):
    account = USERS.get(username)
    if account is None or account["password"] != password:
        context = {"user": None, "error": "Invalid username or password."}
        return templates.TemplateResponse(request, "login.html", context, status_code=401)
    session_id = secrets.token_hex(16)
    ACTIVE_SESSIONS.add(session_id)
    request.session["id"] = session_id
    request.session["user"] = {"username": username, "name": account["name"]}
    request.session["last_seen"] = time.time()
    return RedirectResponse("/dashboard", status_code=302)

@router.get("/dashboard")
def dashboard(request: Request):
    user = current_user(request)
    if not user:
        return RedirectResponse("/login", status_code=302)
    return templates.TemplateResponse(request, "dashboard.html", {"user": user})

@router.get("/logout")
def logout(request: Request):
    ACTIVE_SESSIONS.discard(request.session.get("id"))
    request.session.clear()
    return RedirectResponse("/", status_code=302)

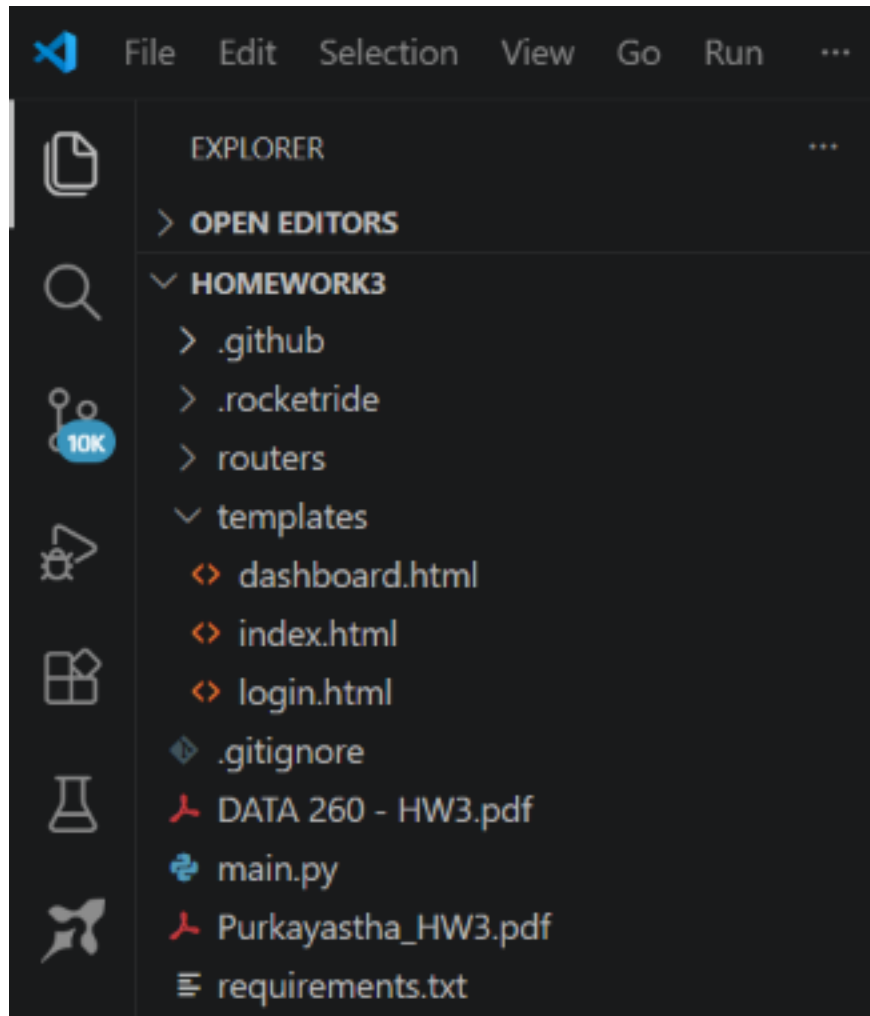
```

Templates directory

```

templates/
  base.html
  dashboard.html
  index.html
  login.html

```



Part 2: three LlamaIndex chunking techniques, retrieval only

Environment

`pip install llama-index llama-index-embeddings-huggingface sentence-transformers faiss-cpu numpy pandas` (pinned in `requirements.txt`). The embedding model is `sentence-transformers/all-MiniLM-L6-v2` from Hugging Face, loaded through `HuggingFaceEmbedding` and run on the CPU. It produces 384-dimensional vectors and reads at most 256 wordpiece tokens of a text; anything longer is cut off before it is embedded, which matters for the semantic splitter below.

Corpus

The graded corpus is 21 plain-text snapshots of public documents about grocery recalls and supply (393,964 bytes, well above the 200 KB minimum): 13 Wikipedia articles (product recalls, FSMA, FSIS, the Peanut Corporation of America, the 2008 Canadian listeriosis outbreak, the 2006 spinach and 2011

German E. coli outbreaks, the 2022 infant formula shortage, the 2021 to 2023 supply chain crisis, cold chain, traceability, food distribution, grocery stores), five FDA pages (recall definitions, FDA 101 on recalls, food recalls for consumers, the FSMA Food Traceability Rule, the FSMA preventive controls rule), a USDA ERS page on food service market segments, 21 CFR Part 7 Subpart C from the eCFR API, and the 100 most recent openFDA food enforcement reports. `fetch_corpus.py` downloads them, strips the HTML, and writes `CORPUS_MANIFEST.json` with the byte size and SHA-256 of every file; `SOURCES.md` lists the URLs and access dates. Tiny Shakespeare is used only as a warm-up and is not committed.

```
$ python fetch_corpus.py
54940 bytes  wikipedia_product_recall.txt
33890 bytes  wikipedia_fda_food_safety_modernization_act.txt
11984 bytes  wikipedia_food_safety_and_inspection_service.txt
...
19297 bytes  ecfr_21_cfr_part_7_subpart_c_recalls.txt
69543 bytes  openfda_food_enforcement_reports.txt
21 files, 393964 bytes total, manifest written to reports\hw03\CORPUS_MANIFEST.json
```

One retrieval-only pipeline per technique

`chunking_compare.py` loads every corpus file as a Document whose only metadata is the file name (excluded from the embedded text), builds the three node parsers, indexes each set of nodes in a `VectorStoreIndex` backed by the default in-memory `SimpleVectorStore`, and runs the same retrieval helper for every question.

```
def build_chunkers(embed_model):
    return {
        "token": TokenTextSplitter(chunk_size=200, chunk_overlap=40),
        "semantic": SemanticSplitterNodeParser(buffer_size=1, breakpoint_percentile_threshold=95,
        embed_model=embed_model),
        "sentence_window": SentenceWindowNodeParser.from_defaults(
            window_size=3, window_metadata_key="window", original_text_metadata_key="original_text"
        ),
    }
```

Token: 200 tokens with a 40 token overlap, chosen so that every chunk fits inside the 256 wordpiece window of the embedding model (the largest token chunk is 218 wordpieces). Semantic: buffer of one sentence and the default 95th percentile breakpoint, using the same MiniLM model to find the boundaries. Sentence window: single sentences as nodes with the three sentences on each side stored in the window metadata, which is excluded from the embedding.

```
def retrieve(technique, index, embed_model, question, k):
    query_vec = np.array(embed_model.get_query_embedding(question["question"]))
    print(f"query embedding: dim={query_vec.shape[0]} first8={np.round(query_vec[:8], 4).tolist()}")
    retriever = index.as_retriever(similarity_top_k=k)
    start = time.perf_counter()
    hits = retriever.retrieve(QueryBundle(query_str=question["question"], embedding=query_vec.
    tolist()))
    latency_ms = (time.perf_counter() - start) * 1000
    doc_vecs = np.array([embed_model.get_text_embedding(hit.node.get_content(
    metadata_mode=MetadataMode.EMBED)) for hit in hits])
    print(f"query vector shape {query_vec.shape}, stacked doc vectors shape {doc_vecs.shape}")
    rows = []
    for rank, (hit, doc_vec) in enumerate(zip(hits, doc_vecs), start=1):
        text = hit.node.get_content(metadata_mode=MetadataMode.NONE)
        window = hit.node.metadata.get("window")
        rows.append({
            "technique": technique,
            "question_id": question["id"],
            "question": question["question"],
            "expected_source": question["expected_source"],
```

```

        "rank": rank,
        "store_score": round(float(hit.score), 4),
        "cosine_sim": round(cosine(query_vec, doc_vec), 4),
        "chunk_len": len(text),
        "preview": text[:160].replace("\n", " "),
        "source_file": hit.node.metadata.get("source_file"),
        "source_hit": hit.node.metadata.get("source_file") == question["expected_source"],
        "contains_answer": contains_all(window or text, question["answer_keywords"]),
        "latency_ms": round(latency_ms, 2),
        "text": text,
        "window": window,
    })
    table = pd.DataFrame(rows)[["rank", "store_score", "cosine_sim", "chunk_len", "preview"]]
    print(table.to_string(index=False))
    print(f"retrieval latency: {latency_ms:.2f} ms")
    return rows

```

The query embedding is computed once and passed to the retriever inside a `QueryBundle`, so the timer only measures the similarity search. The cosine column recomputes the embedding of every returned chunk explicitly and compares it with the query vector; it matches the retriever's `store_score` to four decimals, which confirms that `SimpleVectorStore` scores with cosine similarity. `contains_answer` checks whether the chunk (for the sentence-window index, the stored window, which is what a reader model would receive) contains the answer phrases from `questions.yaml`, matched as whole words. Every row is appended to `raw/retrieval_results.jsonl`, the chunk counts and average lengths go to `raw/chunk_stats.json`, and `summarize_retrieval.py` rebuilds the tables below from those two files.

Warm-up on Tiny Shakespeare

`python chunking_compare.py --warmup` downloads the file, keeps the first 100 KB (the opening of *Coriolanus*) and asks "Why do the citizens want to kill Caius Marcius?". The expected line is "First, you know Caius Marcius is chief enemy to the people."

Technique	Chunks	Avg chunk length (chars)	Top-1 cosine	Mean@5 cosine	Recall@5	Retrieval latency (ms)
token	174	714.4	0.5275	0.4913	0.0	6.18
semantic	59	1694.9	0.5070	0.4581	0.0	2.28
sentence_window	1156	86.5	0.7100	0.6205	1.0	30.43

```
=== Technique: sentence_window | 1156 chunks, avg 86.5 chars, chunking 0.1 s, index build 15.6 s ===
```

```

--- sentence_window | warmup: Why do the citizens want to kill Caius Marcius?
query embedding: dim=384 first8=[0.0009, 0.0291, -0.032, -0.0314, -0.0114, -0.0144, 0.0789, -0.0146]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len preview
1      0.7100      0.7100      76      First Citizen: First,
you know Caius Marcius is chief enemy to the people.
2      0.6639      0.6639      69      Second Citizen:
Would you proceed especially against Caius Marcius?
3      0.6468      0.6468      35
    Messenger: Where's Caius Marcius?
4      0.5498      0.5498      32
    MARCIUS: How lies their battle?
5      0.5319      0.5319     109 If we and Caius Marcius chance to meet, 'Tis sworn
between us we shall ever strike Till one can do no more.
retrieval latency: 30.43 ms

```

Only the sentence-window index put the answer line first; the token and semantic chunks that scored highest were crowd scenes about Marcius that do not contain the line. The script and output format were checked here before the graded run.

Questions

questions.yaml was committed (commit 9f51aaf) before the graded comparison was run. Four of the five questions depend on a single source file.

Id	Question	Expected source	Single source
q1_listeria_plant	Which Maple Leaf Foods facility was the source of the 2008 listeria outbreak in Canada?	wikipedia_2008_canadian_listeria_outbreak.txt	yes
q2_ecoli_sprouts	What was identified as the likely source of the 2011 E. coli O104:H4 outbreak in Germany?	wikipedia_2011_germany_e_coli_outbreak.txt	yes
q3_formula_plant	Which infant formula plant shut down in February 2022 and what bacterium was found there?	wikipedia_2022_infant_formula_shortage.txt	yes
q4_recall_classes	What is the difference between a Class I, Class II and Class III recall?	fda_recalls_background_and_definitions.txt	no
q5_traceability_deadline	What is the compliance date for the FDA Food Traceability Rule?	fda_fsma_food_traceability_rule.txt	yes

The expected answers are in the file. One scoring detail changed after the first run: the answer phrases for q4 were originally the class names, and a plain substring test counted "Class I" inside "Class II" and "Class III", so openFDA records that only say "Classification: Class II" were marked as containing the answer. The check now matches whole phrases and q4 uses the two defining phrases ("serious adverse health consequences", "medically reversible"); the questions and expected answers themselves were not changed.

Results: retrieval quality, five questions, k=5

python chunking_compare.py took 2 min 29 s in total (00:27:52 to 00:30:21 local time, console output in RUN_LOG.txt). The tables come from python summarize_retrieval.py and are also in METRICS.md.

Technique	Chunks	Avg chunk length (chars)	Top-1 cosine	Mean@5 cosine	Recall@5	Mean retrieval latency (ms)
token	557	865.9	0.7368	0.6388	1.0	21.55

Technique	Chunks	Avg chunk length (chars)	Top-1 cosine	Mean@5 cosine	Recall@5	Mean retrieval latency (ms)
semantic	157	2471.9	0.7088	0.5787	1.0	6.41
sentence_window	2618	148.2	0.7836	0.6867	0.8	71.23

Top-1 and Mean@5 are averaged over the five questions; Recall@5 is the share of questions with an answer-bearing chunk in the top 5. Build cost: token 0.9 s chunking and 22.1 s index build, semantic 67.1 s and 7.3 s, sentence window 0.2 s and 33.2 s. Chunk sizes measured in wordpieces of the embedding model:

Technique	Chunks	Wordpieces min / median / max	Chunks longer than the 256 wordpiece window
token	557	56 / 192 / 218	0 (0.0%)
semantic	157	6 / 304 / 5923	86 (54.8%)
sentence_window	2618	4 / 28 / 693	6 (0.2%)

Per question:

Question	Technique	Top-1 cosine	Mean@5	Answer found at rank	Expected source in top-k	Latency (ms)
q1_listeria_plant	token	0.8777	0.6654	1	yes	16.25
q2_ecoli_sprouts	token	0.7582	0.7225	1	yes	19.56
q3_formula_plant	token	0.6724	0.5654	1	yes	23.84
q4_recall_classes	token	0.5988	0.5146	1	yes	28.73
q5_traceability_deadline	token	0.7767	0.7258	1	yes	19.39
q1_listeria_plant	semantic	0.8778	0.619	1	yes	7.61
q2_ecoli_sprouts	semantic	0.7316	0.6486	1	yes	6.16
q3_formula_plant	semantic	0.6494	0.5314	1	yes	5.71
q4_recall_classes	semantic	0.6086	0.4594	1	yes	5.24
q5_traceability_deadline	semantic	0.6765	0.635	2	yes	7.34
q1_listeria_plant	sentence_window	0.8798	0.7002	1	yes	62.58
q2_ecoli_sprouts	sentence_window	0.8702	0.7601	1	yes	71.87
q3_formula_plant	sentence_window	0.6535	0.5695	1	yes	76.67
q4_recall_classes	sentence_window	0.7091	0.645	none	no	73.03
q5_traceability_deadline	sentence_window	0.8054	0.7585	1	yes	72.0

Printed output for the shared query q5

The three blocks below are the console output for "What is the compliance date for the FDA Food Traceability Rule?" (query embedding with dimension and first eight values, the vector shapes, the rank table and the latency), one per technique.

```
--- token | q5_traceability_deadline: What is the compliance date for the FDA Food Traceability Rule?
```

query embedding: dim=384 first8=[-0.05, -0.0272, -0.0422, 0.0347, 0.0445, -0.0034, -0.0477, -0.001]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len

rank	store_score	cosine_sim	chunk_len	document
1	0.7767	0.7767	940	Food Traceability Final Rule requires entities to share information with other entities in their supply chain, the most effective and
2	0.7257	0.7257	1048	The new requirements identified in the final rule will allow for faster identification and rapid removal of potentially contaminated
3	0.7145	0.7145	951	compliance. This meeting was part of a series of engagements being held in accordance with a directive from Congress in the
4	0.7123	0.7123	890	Foundation held a virtual public meeting to hear insights on the Food Traceability Rule from both FDA leadership and other experts fr
5	0.7000	0.7000	306	Safety Rule and the Food Traceability Rule Risk-ranking Model Results Tool Tracking and Tracing of Food Discussion Paper: Identifying

retrieval latency: 19.39 ms

--- semantic | q5_traceability_deadline: What is the compliance date for the FDA Food Traceability Rule?
query embedding: dim=384 first8=[-0.05, -0.0272, -0.0422, 0.0347, 0.0445, -0.0034, -0.0477, -0.001]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len

rank	store_score	cosine_sim	chunk_len	document
1	0.6765	0.6765	2400	Federal Register Notice Docket Folder FDA-2011-N-0920 provides the full text of the Rule Questions & Answers Fact Sheet Generally,
2	0.6491	0.6491	4525	What Critical Tracking Events (CTEs) do you conduct? What Key Data Elements (KDEs) do you already maintain? What additional KDEs
3	0.6375	0.6375	9827	The exemptions are listed in § 1.1305 of the final rule. A tool is available to help stakeholders determine whether an exemption m
4	0.6095	0.6095	1926	The agency estimated that it would require at least 1,000 more inspectors and \$1.4 billion over the following five years, with uncert
5	0.6024	0.6024	3709	What's New The FDA held a public meeting on June 15, 2026, to give the public an opportunity to share information on continued implem

retrieval latency: 7.34 ms

--- sentence_window | q5_traceability_deadline: What is the compliance date for the FDA Food Traceability Rule?
query embedding: dim=384 first8=[-0.05, -0.0272, -0.0422, 0.0347, 0.0445, -0.0034, -0.0477, -0.001]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len

rank	store_score	cosine_sim	chunk_len	document
1	0.8054	0.8054	148	The original compliance date for all persons subject to the recordkeeping requirements of the Food Traceability Rule was
2	0.7845	0.7845	251	(October 7, 2024) Contact Us If you have questions on the Food Traceability Rule, please submit your inquiry via FDA's Food and Cos
3	0.7840	0.7840	288	Compliance Date Because the Food Traceability Final Rule requires entities to share information with other entities in their supply c
4	0.7191	0.7191	231	Subsequently, the Continuing Appropriations, Agriculture, Legislative Branch, Military Construction and Veterans Affairs, and Extens
5	0.6994	0.6994	516	Exemptions to the Food Traceability Rule Exenciones a la Norma de trazabilidad de alimentos Modified Requirements, Exemptions, and W

retrieval latency: 72.00 ms

```

TERMINAL powershell

PS C:\Users\Poushall\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> python chunking_compare.py
21 documents, 388095 characters, k=5
=== Technique: token | 557 chunks, avg 865.9 chars, chunking 0.9 s, index build 22.1 s ===
...
=== Technique: sentence_window | 2618 chunks, avg 148.2 chars, chunking 0.2 s, index build 33.2 s ===
--- sentence_window | q1_listeriosis_plant: Which Maple Leaf Foods facility was the source of the 2008 listeriosis outbreak in Canada?
query embedding: dim=384 first8=[-0.0228, -0.0423, 0.0178, 0.0237, 0.0409, 0.0177, -0.0467, 0.0291]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len
preview
1 0.8798 0.8798 163 The 2008 Canadian listeriosis outbreak was a widespread outbreak of listeriosis in Canada linked to cold cuts from
a Maple Leaf Foods plant in Toronto , Ontario
2 0.7264 0.7264 116 August: Maple Leaf Foods voluntarily recalled a number of meat and del
i products after an outbreak of listeriosis .
3 0.7262 0.7262 107 The outbreak originated from lines 8 and 9 of the Maple Leaf
Foods Bortor Road facility (Establishment No.
4 0.6206 0.6206 133 Officials from Maple Leaf believe that the outbreak originated sometime in July 2008 on
line 8 or line 9 of the North York facility.
5 0.5482 0.5482 70 November: Kenny's farmho
use 4 cheeses were recalled due to listeria.
retrieval latency: 62.58 ms
--- sentence_window | q2_e_coli_sprouts: What was identified as the likely source of the 2011 E. coli O104:H4 outbreak in Germany?
query embedding: dim=384 first8=[0.011, 0.0307, -0.0120, 0.057, 0.0586, 0.0652, -0.0644, 0.0391]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len
preview
1 0.8702 0.8702 153 A novel strain of Escherichia coli O104:H4 bacteria caused a serious outbreak of foodborne illness focused
in northern Germany in May through June 2011.
2 0.8170 0.8170 266 A joint risk-assessment by EFSA / ECDC , issued 29 June 2011, made a connection between the German outbreak and a
HUS outbreak in the Bordeaux area of France, f
3 0.7096 0.7096 189 Cause The outbreak was caused by a strain of E. coli of the serotype O104:H4, that was unusual for having charact
eristics of both enteroaggregative E. coli and
4 0.7072 0.7072 224 On 4 June, German and EU officials had allegedly been examining data that indicated an open catering event at a re
staurant in Lübeck , Germany, was a possible s
5 0.6964 0.6964 198 The most substantial outbreak outside of Germany was in Bordeaux, France where 15 cases of EHEC gastroenteritis we
re associated with the same strain of E. coli
retrieval latency: 71.87 ms
--- sentence_window | q3_formula_plant: Which infant formula plant shut down in February 2022 and what bacterium was found there?
query embedding: dim=384 first8=[-0.060, 0.011, -0.0501, 0.0286, 0.0593, 0.0438, -0.1193, 0.0878]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len
preview
1 0.6535 0.6535 369 Recall and plant shutdown The problem was exacerbated when Abbott Labs , one of three major formula producers in
the United States, shut down production at its
2 0.5654 0.5654 284 In 2022, the United States experienced a severe shortage of infant formula as a result of the 2021-2022 global sup
ply chain crisis compounded by a large scale p
3 0.5626 0.5626 98 The bacterium which sickened the infants, Cronobacte
r sakazakii , was found at the Sturgis plant.
4 0.5356 0.5356 207 On June 10, the FDA stated that it was investigating reports that nine children died after consuming infant formul
a produced in the Sturgis plant, seven more th
5 0.5306 0.5306 385 An investigation by ABC News of FDA inspection reports dating back five years found that in that time, three other
major formula manufacturers - Mead Johnson (o
retrieval latency: 76.67 ms
--- sentence_window | q4_recall_classes: What is the difference between a Class I, Class II and Class III recall?
query embedding: dim=384 first8=[-0.0588, -0.0601, -0.0611, -0.0322, 0.0088, -0.0173, 0.0098, 0.0057]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len
preview
1 0.7091 0.7091 259 (b) On the basis of this determination, the Food and Drug Administration will assign the recall a classification,
i.e., Class I, Class II, or Class III, to indi
2 0.6336 0.6336 326 Classification: Class III Status: Ongoing Recall initiated: 20260715, report date: 20260826 Product quantity: 21,4
00 units (total) Distribution: AK, AL, AR, AZ,
3 0.6336 0.6336 326 Classification: Class III Status: Ongoing Recall initiated: 20260715, report date: 20260826 Product quantity: 21,4
00 units (total) Distribution: AK, AL, AR, AZ,
4 0.6336 0.6336 326 Classification: Class III Status: Ongoing Recall initiated: 20260715, report date: 20260826 Product quantity: 21,4
00 units (total) Distribution: AK, AL, AR, AZ,
5 0.6152 0.6152 251 Classification: Class III Status: Ongoing Recall initiated: 20260728, report date: 20260819 Product quantity: 80 c
ases Distribution: Product is sold to independ
retrieval latency: 73.03 ms
--- sentence_window | q5_traceability_deadline: What is the compliance date for the FDA Food Traceability Rule?
query embedding: dim=384 first8=[-0.05, -0.0272, -0.0422, 0.0347, 0.0445, -0.0034, -0.0477, -0.0001]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len
preview
1 0.8054 0.8054 148 The original compliance date for all persons subject to the recordkeeping requirements of the Food Tra
ceability Rule was Tuesday, January 20, 2026.
2 0.7845 0.7845 251 (October 7, 2024) Contact Us If you have questions on the Food Traceability Rule, please submit your inquiry via
FDA's Food and Cosmetics Information Center (
3 0.7840 0.7840 288 Compliance Date Because the Food Traceability Final Rule requires entities to share information with other entitie
s in their supply chain, the most effective an
4 0.7191 0.7191 231 Subsequently, the Continuing Appropriations, Agriculture, Legislative Branch, Military Construction and Veterans
Affairs, and Extensions Act of 2026 directed
5 0.6994 0.6994 516 Exemptions to the Food Traceability Rule Exenciones a la Norma de trazabilidad de alimentos Modified Requirement
s, Exemptions, and Waivers Modified Requirement
retrieval latency: 72.00 ms
wrote C:\Users\Poushall\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export\reports\hw03\raw\retrieval_results.jsonl and C:\Users\Poushall\Documents\DATA
-260_Agentic\Homework1_AWS_Docker_export\reports\hw03\raw\chunk_stats.json

```

Confidently scored retrievals that do not contain the answer

Question	Technique	Rank	Cosine	Source file	What the chunk says
q5_traceability_deadline	semantic	1	0.6765	fda_fsma_preventive_controls_human_food.txt	A 2,400 character chunk about the FSMA preventive controls rule: its Federal Register docket, "requires food facilities to have a food safety plan", "Compliance dates are staggered, based on the size of the business". No mention of the traceability rule or of January 20, 2026.
q4_recall_classes	sentence_window	1	0.7091	ecfr_21_cfr_part_7_subpart_c_recalls.txt	21 CFR 7.41(b): "the Food and Drug Administration will assign the recall a classification, i.e., Class I, Class II, or Class III, to indicate the relative degree of health hazard". It names the classes but never defines them.
q4_recall_classes	sentence_window	2 to 5	0.6336 to 0.6152	openfda_food_enforcement_reports.txt	Lines such as "Classification: Class II" from individual enforcement records.

Question	Technique	Rank	Cosine	Source file	What the chunk says
q5_traceability_deadline	token	2 to 5	0.7257 to 0.7000	fda_fsma_food_traceability_rule.txt	Paragraphs about the traceability rule (benefits, public meetings, tools) without the compliance date.

The sentence-window output for q4 shows the second case:

```

--- sentence_window | q4_recall_classes: What is the difference between a Class I, Class II and Class III recall?
query embedding: dim=384 first8=[-0.0588, -0.0601, -0.0611, -0.0322, 0.0088, -0.0173, 0.0098, 0.0057]
query vector shape (384,), stacked doc vectors shape (5, 384)
rank store_score cosine_sim chunk_len

1      0.7091      0.7091      259 (b) On the basis of this determination, the Food and Drug
Administration will assign the recall a classification, i.e., Class I, Clas
2      0.6336      0.6336      326 Classification: Class III Status: Ongoing Recall
initiated: 20260715, report date: 20260826 Product quantity: 21,400 units (total) Di
3      0.6336      0.6336      326 Classification: Class III Status: Ongoing Recall
initiated: 20260715, report date: 20260826 Product quantity: 21,400 units (total) Di
4      0.6336      0.6336      326 Classification: Class III Status: Ongoing Recall
initiated: 20260715, report date: 20260826 Product quantity: 21,400 units (total) Di
5      0.6152      0.6152      251 Classification: Class III Status: Ongoing Recall
initiated: 20260728, report date: 20260819 Product quantity: 80 cases Distribution:
retrieval latency: 73.03 ms

```

Why the embedding considered them similar: the q5 semantic miss shares almost all of the question's vocabulary (FDA, FSMA, rule, compliance date), and the model cannot tell the preventive controls rule from the traceability rule when the one distinguishing word, "traceability", is absent from the part of the chunk that was embedded. The traceability chunk that holds the date is 4,525 characters long, the date sits far past the 256 wordpiece cut-off, and its embedded head is about critical tracking events and key data elements, so it landed at rank 2 (0.6491). The q4 sentence-window miss is the opposite case: a single sentence that lists "Class I, Class II, or Class III" next to "recall" and "health hazard" is an almost perfect lexical match for a question that lists the same three names, while the actual definitions on the FDA page are written as a label line ("Class I recall:") followed by the definition on the next line, so the sentence splitter stores the label and the definition as separate short nodes and neither one alone scores well.

Observations

The sentence-window index has the highest top-1 and mean@5 cosines and put the exact answer sentence first for four of the five questions (for example "The original compliance date for all persons subject to the recordkeeping requirements of the Food Traceability Rule was Tuesday, January 20, 2026" at 0.8054, and the Bartor Road sentence at 0.8798). A one-sentence node is the closest thing to a one-sentence question, so the scores are high, and the three-sentence window carries the context that the node itself lacks. The cost is 2,618 vectors instead of 557, which makes the linear search about three times slower than the token index (71 ms against 22 ms), and the q4 failure, where a sentence that names the topic beats the sentences that answer the question. Token chunks of 200 tokens with a

40 token overlap were the most robust: the answer was at rank 1 for every question, every chunk fits inside the model window, and each hit brings roughly 870 characters of surrounding text with it. Its cosines are lower than the sentence-window ones because each chunk also contains material unrelated to the query.

The semantic splitter found coherent boundaries (its q1 chunk is exactly the "Origin of the outbreak" section) but produced chunks that average 2,472 characters, and 55% of them are longer than the 256 wordpieces MiniLM reads, so most of a long chunk never influences its embedding. That is why it has the lowest mean@5 (0.5787), why the q5 answer dropped to rank 2, and why one retrieved chunk is 9,827 characters long. It is the fastest at query time (157 vectors, 6.4 ms) and the slowest to build (67 s of sentence embeddings for boundary detection). The best technique does change with the query: sentence window had the highest top-1 cosine on q1, q2, q4 and q5 and token on q3, but on q4 only token and semantic actually returned the definitions. On the Tiny Shakespeare warm-up the sentence-window index was the only one to put the answer line first.

Conclusion

For this corpus I judge token chunking (200 tokens, 40 overlap) the best of the three: it is the only technique that returned a chunk containing the answer at rank 1 for all five questions, none of its 557 chunks exceeds the embedding window, and each hit carries enough context to answer from, at 22 ms per search. Sentence-window chunking scores highest on cosine (0.7836 top-1, 0.6867 mean@5) and is the better choice when a question maps to a single sentence, but it missed the recall class definitions entirely and needs three times the search time plus the window post-processing step. Semantic chunking gave the lowest scores here because its oversized chunks are truncated by the 256 token model; it would need a smaller breakpoint threshold or a longer-context embedding model to compete.

Verification

`python verify_hw03.py` (also make `verify-hw03`) starts the app on port 8571 with a 5 second idle timeout, checks the four routes, the three cookie attributes, the logout and idle replays, the corpus size and SHA-256 hashes, the five questions, the raw results for all three techniques, and that `summarize_retrieval.py` recomputes the tables, then writes `reports/hw03/verification.json`.

```
$ date
2026-09-21T00:34:15-07:00
$ python verify_hw03.py
PASS home_page_shows_login_link: GET / returned 200
PASS invalid_login_shows_alert: POST /login with a wrong password returned 401
PASS login_redirects_to_dashboard: returned 302, Location=/dashboard
PASS session_cookie_has_three_flags: path=/; Max-Age=5; httponly; samesite=lax; secure
PASS dashboard_with_session: returned 200
PASS dashboard_without_session_redirects: returned 302, Location=/login
PASS logout_redirects_home: returned 302
PASS logged_out_cookie_rejected: old cookie on /dashboard returned 302
PASS idle_session_expires: after 6 s idle /dashboard returned 302
PASS corpus_at_least_200_kb: 21 files, 393964 bytes
PASS corpus_hashes_match_manifest: all hashes match
PASS five_questions_with_expected_answers: 5 questions
PASS at_least_two_single_source_questions: single-source: ['q1_listeria_plant',
'q2_ecoli_sprouts', 'q3_formula_plant', 'q5_traceability_deadline']
PASS raw_results_cover_three_techniques: 75 rows
PASS raw_rows_have_required_fields: store_score, cosine_sim, chunk_len, latency_ms, preview
PASS summary_script_recomputes_tables: exit code 0
```

```
PASS report_files_present: all files present
wrote C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export\reports\hw03\
verification.json (passed=True)
exit code 0
```

TERMINAL powershell

```
PS C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export> python verify_hw03.py
PASS home_page_shows_login_link: GET / returned 200
PASS invalid_login_shows_alert: POST /login with a wrong password returned 401
PASS login_redirects_to_dashboard: returned 302, Location=/dashboard
PASS session_cookie_has_three_flags: path=/; Max-Age=5; httponly; samesite=lax; secure
PASS dashboard_with_session: returned 200
PASS dashboard_without_session_redirects: returned 302, Location=/login
PASS logout_redirects_home: returned 302
PASS logged_out_cookie_rejected: old cookie on /dashboard returned 302
PASS idle_session_expires: after 6 s idle /dashboard returned 302
PASS corpus_at_least_200_kb: 21 files, 393964 bytes
PASS corpus_hashes_match_manifest: all hashes match
PASS five_questions_with_expected_answers: 5 questions
PASS at_least_two_single_source_questions: single-source: ['q1_listeria_plant', 'q2_ecoli_sprouts', 'q3_formula_plant', 'q5_traceability_deadline']
PASS raw_results_cover_three_techniques: 75 rows
PASS raw_rows_have_required_fields: store_score, cosine_sim, chunk_len, latency_ms, preview
PASS summary_script_recomputes_tables: exit code 0
PASS report_files_present: all files present
wrote C:\Users\Poushali\Documents\DATA-260_Agentic\Homework1_AWS_Docker_export\reports\hw03\verification.json (passed=True)
```

AI-use statement

Same text as reports/hw03/AI_USE.md.

1. I used an AI assistant for debugging and for preparing the report document (report.md / report.pdf). The debugging help covered the login flow (why a replayed cookie still opened the dashboard, why the smoke test looked slow), the Wikipedia rate limit that broke the first corpus download, a console encoding error on a narrow no-break space in the FDA text, and the reportlab layout. I wrote and ran the application code, the chunking comparison, the corpus fetch script and the smoke test myself, chose the corpus sources, wrote the five questions before running the comparison, reviewed every output file, and made the Canvas submission.
2. One thing I verified myself was the logout. The first version of the logout route only cleared the session, so the browser dropped the cookie, but a copy of the signed cookie stayed valid until its Max-Age ran out: replaying it with curl after /logout still returned 200 OK from /dashboard. A second unsuitable result was my own answer check in the retrieval script: it counted "Class I" as present inside "Class II" and "Class III", so openFDA records that only say "Classification: Class II" were reported as containing the definition of the recall classes. I also checked that the store_score returned by the LlamaIndex retriever is the cosine similarity by recomputing the chunk embeddings explicitly.
3. I found the replay problem with the curl sequence in RUN_LOG.txt (log in,

log out, send the old cookie again) against a server started with `IDLE_TIMEOUT_SECONDS=20`. An earlier check with a 3 second timeout had hidden it, because every request to `localhost` on this laptop takes about 2 seconds (IPv6 is tried first), so the cookie had already expired by the time it was replayed; timing requests against `localhost` and `127.0.0.1` showed the difference (2.1 s versus 0.015 s). I found the answer-check problem by reading the per-question rows for q4: four openFDA chunks at ranks 2 to 5 were flagged as containing the answer although their text is a single classification line. The cosine check is the `cosine_sim` column next to `store_score` in every table: the values match to four decimals.

4. `routers/auth.py` now keeps a set of active session ids. Login creates a random id, stores it in the session and in the set; logout removes it from the set before clearing the session; `current_user()` rejects any cookie whose id is not in the set or whose `last_seen` is older than the idle timeout, so a replayed cookie is refused with a redirect to `/login` even though its signature is still valid. `verify_hw03.py` and the run log use `127.0.0.1` and a 5 second timeout, so the checks measure the application and not the name lookup. The answer check now matches whole phrases with a regular expression, and q4 uses the two defining phrases from the FDA page; the questions and expected answers in `questions.yaml` were not changed. With that fix the sentence-window index shows a real miss on q4 (Recall@5 of 0.8), which is discussed in `METRICS.md`.